The appetizer classifier: pairs of a type with a map to B are the families of types over B

The Univalent Foundations Program

hott-book:equivalences.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'thm:nobject-classifier-appetizer' (source lines 821-826, Theorem 4.8.3); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. Two context spans are retained, namely the two lemmas the proof invokes by label; with them every used label is defined inside the package.

Editorial note. The retained counter values (chapter 4, section 8, theorem offset 0) make the first retained context print as Theorem 4.8.1, the second as Theorem 4.8.2 and the counted theorem as Theorem 4.8.3, matching the original book numbering taken from the label-number dump.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-task copy inside this task directory and referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: the appetizer version of the object classifier (Theorem 4.8.3).

Allowed standard prerequisites (editorial).

- Fiber of a map. hfiber f b is the type of pairs (a, f a = b).. It is the value of the map chi at a pair ((A,f),b).
- Univalence and its computation rule. An equivalence e gives a path ua(e), and transport along ua(e) is precomposition with the inverse of e.. It is the computation at source line 848.
- Transport in a function type. Transport along a path in a type family of function types acts by precomposition.. It is the identity eq:transport-arrow used at source line 848.

Source reference. Original equation 2.9.4: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1324–1324. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original theorem 2.7.2: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1853–1853. Original result cited by the retained

text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Lemma 4.8.1. *For any type family $B : A \to \mathcal{U}$, the fiber of $\mathsf{pr}_1 : \sum_{(x:A)} B(x) \to A$ over $a : A$ is equivalent to $B(a)$:*

$$\mathsf{fib}_{\mathsf{pr}_1}(a) \simeq B(a)$$

Lemma 4.8.2. *For any function $f : A \to B$, we have $A \simeq \sum_{(b:B)} \mathsf{fib}_f(b)$.*

Theorem 4.8.3. *For any type B there is an equivalence*

$$\chi : \Big(\sum_{A:\mathcal{U}} (A \to B)\Big) \simeq (B \to \mathcal{U}).$$

Proof. We have to construct quasi-inverses

$$\chi : \Big(\sum_{A:\mathcal{U}} (A \to B)\Big) \to B \to \mathcal{U}$$
$$\psi : (B \to \mathcal{U}) \to \Big(\sum_{A:\mathcal{U}} (A \to B)\Big).$$

We define χ by $\chi((A, f), b) :\equiv \mathsf{fib}_f(b)$, and ψ by $\psi(P) :\equiv \Big((\sum_{(b:B)} P(b)), \mathsf{pr}_1\Big)$. Now we have to verify that $\chi \circ \psi \sim \mathsf{id}$ and that $\psi \circ \chi \sim \mathsf{id}$.

(i) Let $P : B \to \mathcal{U}$. By Lemma 4.8.1, $\mathsf{fib}_{\mathsf{pr}_1}(b) \simeq P(b)$ for any $b : B$, so it follows immediately that $P \sim \chi(\psi(P))$.

(ii) Let $f : A \to B$ be a function. We have to find a path

$$\Big(\textstyle\sum_{(b:B)} \mathsf{fib}_f(b),\ \mathsf{pr}_1\Big) = (A, f).$$

First note that by Lemma 4.8.2, we have $e : \sum_{(b:B)} \mathsf{fib}_f(b) \simeq A$ with $e(b, a, p) :\equiv a$ and $e^{-1}(a) :\equiv (f(a), a, \mathsf{refl}_{f(a)})$. By original theorem 2.7.2, it remains to show $(\mathsf{ua}(e))_*(\mathsf{pr}_1) = f$. But by the computation rule for univalence and (2.9.4), we have $(\mathsf{ua}(e))_*(\mathsf{pr}_1) = \mathsf{pr}_1 \circ e^{-1}$, and the definition of e^{-1} immediately yields $\mathsf{pr}_1 \circ e^{-1} \equiv f$. □